Terraform workspaces

1.create isolated workspaces for different environments like dev,prod,qa (workspaces)

2.why we create workspace = maintain a state file separately for different environments

3. Commands :

- terraform workspace show (current WS)
- terraform workspace list (shows all WS)
- terraform workspace new dev (Create new WS)
- terraform workspace delete dev (delete old WS)
- terraform workspace select prod (switch workspace)

Practicals:

- terraform workspace show

```
[root@ip-172-31-24-233 ec2]# terraform workspace show
dev
```

- terraform workspace list

```
[root@ip-172-31-24-233 ec2]# terraform workspace list
  default
* dev
  prod
```

- terraform workspace delete prod

```
[root@ip-172-31-24-233 ec2]# terraform workspace list
  default
* dev
  prod

[root@ip-172-31-24-233 ec2]# terraform workspace delete prod
Deleted workspace "prod"!
[root@ip-172-31-24-233 ec2]# terraform workspace list
  default
* dev
```

- terraform workspace new prod

```
[root@ip-172-31-24-233 ec2]# terraform workspace new prod
Created and switched to workspace "prod"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
[root@ip-172-31-24-233 ec2]# terraform workspace show
prod
```

Note : If you're using an S3 backend

```
terraform {

    required_providers {

        aws = {

            source      = "hashicorp/aws"

            version     = "~> 5.0"

        }

    }
```

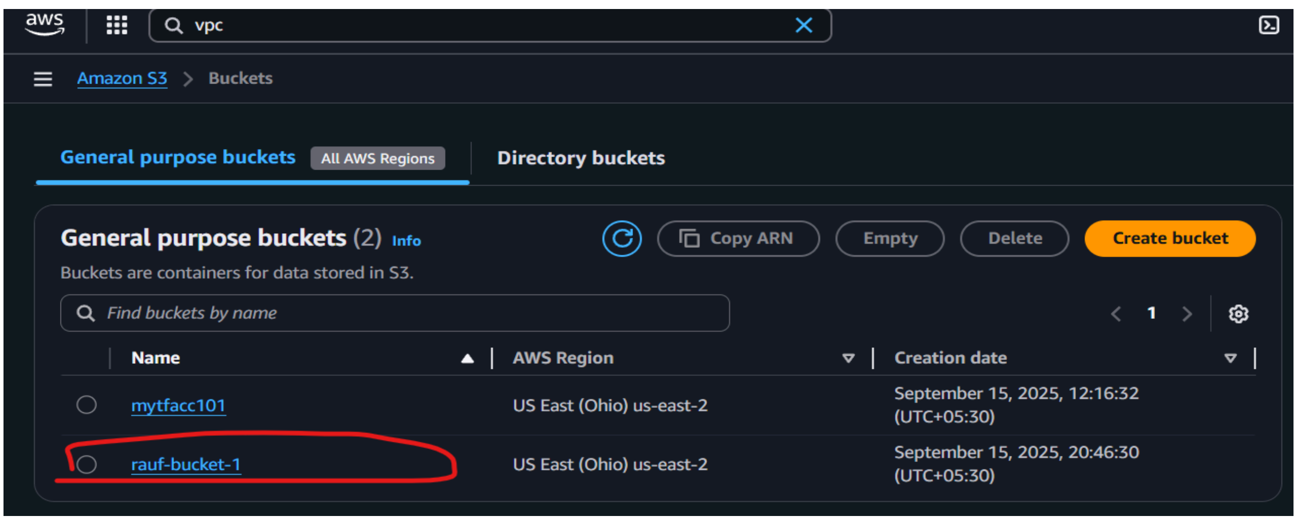
```
    backend "s3" {
        bucket      ="rauf-bucket-1"   #bucket name
        key         = "dev/terraform.tfstate"   #inside the bucket path
        region      = "us-east-2"
    }
}
```

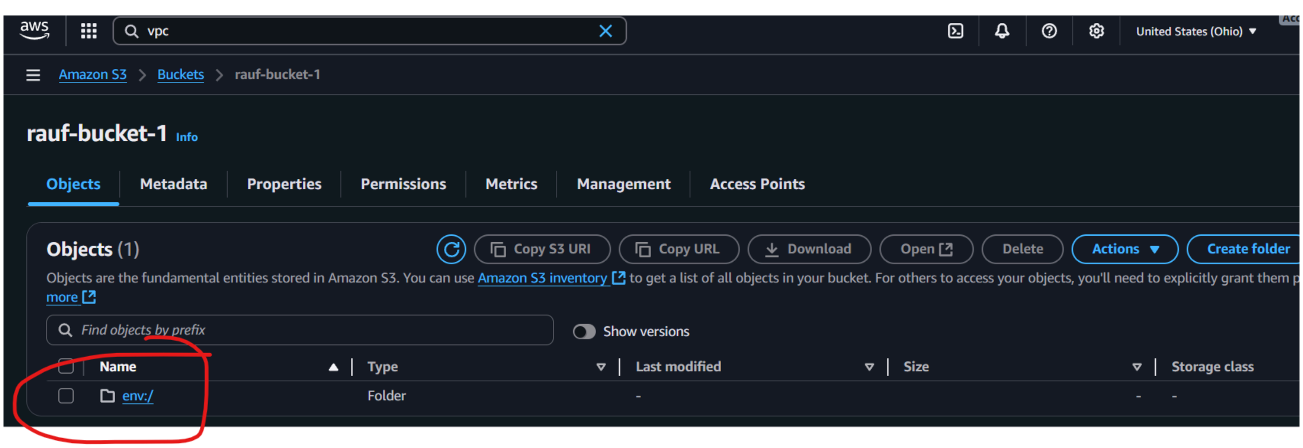
Terraform will automatically create separate state paths per workspace when I run the terraform apply command.

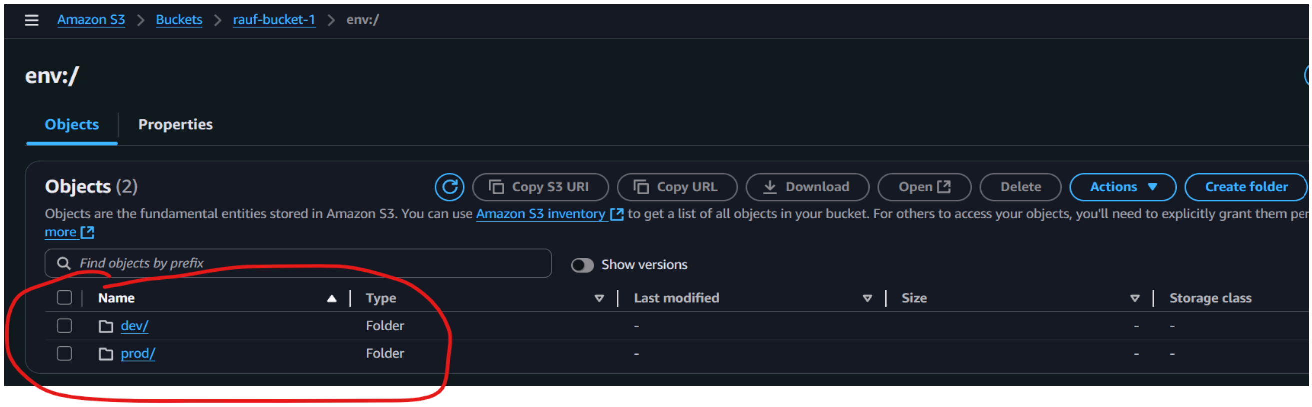
```
[root@ip-172-31-24-233 ec2]# terraform apply --auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.east_server will be created
  + resource "aws_instance" "east_server" {
      + ami                                  = "ami-0634f3c109dcdc659"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.large"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = (known after apply)
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
```

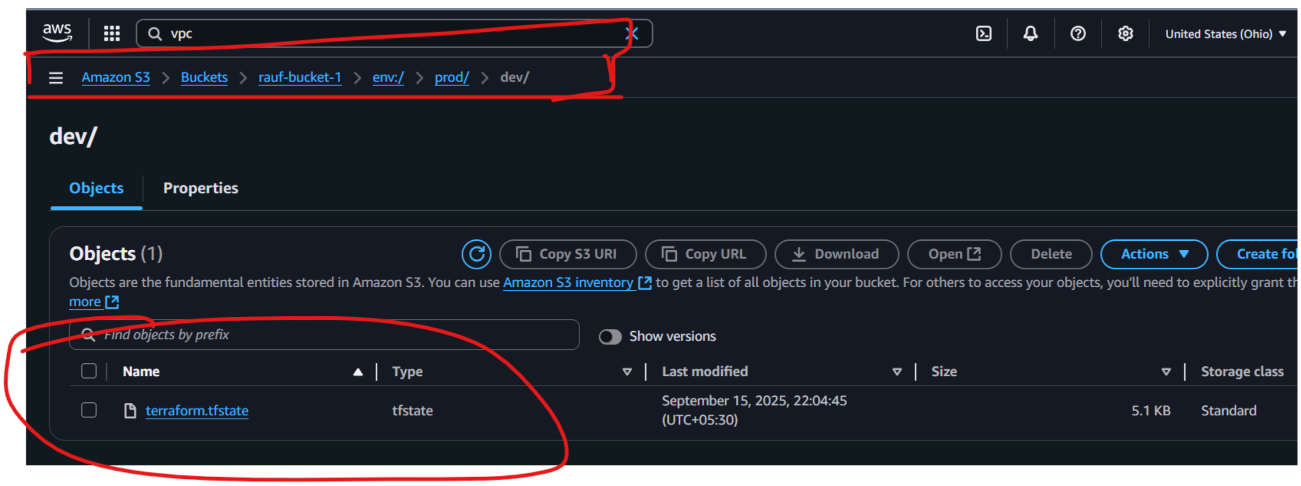
```
Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + east_server_ips = {
      + private_ip = (known after apply)
      + public_ip  = (known after apply)
    }
aws_instance.east_server: Creating...
aws_instance.east_server: Still creating... [00m10s elapsed]
aws_instance.east_server: Creation complete after 12s [id=i-0464f8288243b4bc5]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

east_server_ips = {
  "private_ip" = "172.31.16.101"
  "public_ip" = "3.23.88.214"
}
[root@ip-172-31-24-233 ec2]#
```

Let we check the s3 bucket !









Same for all the workspaces and manage the individual state file for different environments.